PART 5 · BUILDING OPENCLAW APPS

Chapter 57
Extend Your Claw with MCP

Detailed Notes & MCQs — Roman Urdu

✍️ Imran Hassan

5	15	MCP
LESSONS	MCQS	MODEL CONTEXT PROTOCOL

Notes — Roman Urdu

1

Your Agent Needs Hands

Agent Ko Haath Chahiye

Background Story

- James ne Chapter 56 mein ek kaam karta AI Employee banaya tha
- Uska agent messages handle karta, cheezein yaad rakhta, schedules manage karta tha
- Lekin jab student ne pucha ke usne pichla lesson complete kiya ya nahi — James ko pata chala ke agent yeh kaam **nahi** kar sakta

Agent Ki Buniyadi Limitation

> *"Training data gives knowledge: what timezones exist, how offsets work, when daylight saving starts in theory. A tool gives ability: what the actual time is, right now, on a live system."*

- Agent **jaanta hai** ke timezones kya hote hain — lekin **abhi live time** nahi bata sakta bina tool ke
- Agent **samajhta hai** databases kaise kaam karte hain — lekin actual query nahi maar sakta
- Agent routing algorithms explain kar sakta hai — lekin container ka customs status check nahi kar sakta

Yeh cheezein agent bina tool ke NAHI kar sakta: Student progress check karna · Submissions verify karna · Learner ka naam system mein dhundhna · Notifications bhejna · API ping karna

Consumer vs Producer Framework

Type	Matlab	Zimmedari
Consumer	Kisi aur ka MCP server use karna	Sirf configure karo
Producer	Apna khud ka MCP server banana	Design karo + banao

- Chapter 56 mein James **consumer** tha — usne `mcp-server-time` install kiya tha

- Chapter 57 mein James **producer** banega — apna server khud banana seekhega
- Protocol dono taraf **same rehta hai** — sirf zimmedari badal jaati hai

Skill-Driven Build Pattern

Skill Install → CLAUDE.md Configure → Requirements Describe → Claude Code Builds → Verify

> *"You own the requirements. The agent owns the code."*

2 Set Up the Workshop

Workshop Setup Karo

Step 1: Khali Directory Banao

```
mkdir tutorclaw-mcp cd tutorclaw-mcp
```

- Bilkul **khali directory** se shuru karo — koi `package.json`, koi boilerplate nahi
- Claude Code baad mein sab project files **khud generate** karega
- `.claude/skills/` directory build process mein survive karti hai

Step 2: mcp-builder Skill Install Karo

```
npx skills add https://github.com/anthropics/skills --skill mcp-builder
```

Yeh kya add karta hai:

- `.claude/skills/mcp-builder/` directory
- `SKILL.md` file — core component, isme pura MCP knowledge hai
- `reference/` aur `scripts/` folders

Chapter 56 se farq: Wahan skill Claude Code ko MCP *use* karna sikhati thi. Yahan skill Claude Code ko MCP server *banana* sikhati hai — jaise general contractor vs licensed electrician.

Step 3: CLAUDE.md Banao

```
# TutorClaw MCP Server ## Rules - Use the mcp-builder skill for all MCP work - Use uv for Python projects - Follow spec-first development: discuss requirements, output spec, then build - Use flat Annotated parameters on tools, not Pydantic BaseModel wrappers ## After every build 1. Write and run tests. Fix any failures before continuing. 2. Start the server and confirm it boots without errors. 3. Make real HTTP tool calls against the running server. 4. Kill the server process. 5. Report results.
```

Rule	Kyun?
mcp-builder skill use karo	Claude Code generic/outdated knowledge use na kare
uv use karo Python ke liye	pip/poetry se faster, reproducible environments
Spec-first development	Pehle requirements agree karo — rework se bacho

Step 4: Language Choose Karo

- Python — FastMCP ke saath, decorator-based clean API
- TypeScript — TypeScript MCP SDK ke saath
- Dono equally valid — sirf syntax alag, concepts **same**

Final Project Structure

```
tutorclaw-mcp/ ├── CLAUDE.md └── .claude/ └── skills/ └── mcp-builder/ ├── SKILL.md ├── reference/ └── scripts/
```

Abhi koi actual code nahi — sirf configuration aur skill. Yahi pattern Chapter 14 jaise hai: pehle foundation configure karo.

3 Choose Your Wire

Transport Protocol Choose Karo

Transport Protocol Kya Hota Hai?

MCP server aur agent ke beech **communication ka tarika** — decide karta hai ke data kaise travel karega.

Teen Transport Options

Transport	Kab Use Karo	Network?
stdio	Local-only operation	Nahi — same machine
SSE (Server-Sent Events)	Network accessible, legacy use	Haan
Streamable HTTP	Network deployment — Recommended	Haan

Stateful vs Stateless

Type	Matlab	Problem
Stateful	Server sessions yaad rakhta hai	Session cleanup, memory leaks!
Stateless	Har request mein puri info hoti hai	Koi problem nahi ✓

⚠️ **Emma ki Cautionary Tale:** Usne inventory system ke liye stateful transport choose kiya — timezone-related memory leaks aur mid-request session expirations ki wajah se raat **2 baje** debugging karni padi!

> *"Every label has the complete destination address. The warehouse does not need to remember which dock you used last time." — Shipping Label Analogy for Stateless*

Teen Sawaalon Se Decision Guide Karo

- Kya server sirf local machine par chalega ya network par?

- Kya requests ke beech session tracking zaroori hai?
- Kya MCP server ko conversation memory maintain karni hai, ya agent handle karega?

James Ka Final Decision

Streamable HTTP · Stateless · Port 8000

CLAUDE.md mein rule add kiya: `Use streamable-http stateless transport on port 8000`

4

Spec and Build Your First Tool

Pehla Tool Banao — Describe-Steer-Verify Cycle

James Ka Pehla Requirement

Tool: `register_learner` — naye student ko register karna — sab se pehla zaroori kaam

4-Step Describe-Steer-Verify Cycle

Step 1 — Describe (Claude Code Ko Batao)

- Batao: **Kya** server karta hai
- Batao: **Kya** inputs accept karta hai aur outputs return karta hai
- Explicitly kaho: **spec pehle banao**, phir code

```
"Build an MCP server called 'tutorclaw' with a tool to register new learners.
The tool should accept a learner's name. Tell me the spec first before
writing any code."
```

Implementation details na batao — Python code, file paths, framework choices — Claude Code khud decide karega.

Step 2 — Review the Spec

Claude Code yeh generate karta hai:

- Tool name — lowercase, jaise `register_learner`
- **Tool description** — sabse zaroori! Agent isi se decide karta hai ke kab tool call kare
- Input parameters — required / optional
- Output format — success aur error dono ke liye

Buri Description ❌	Achi Description ✓
"Registers stuff"	"Registers a new learner in the TutorClaw system. Call this tool when a user wants to enroll a new student, not for updating existing learner information."

Step 3 — Steer (Guide Karo Agar Zaroori Ho)

- Description sharp banwao
- Optional parameters add karwao (jaise email field)
- Output structure adjust karwao (jaise timestamp include karo)
- Multiple iterations **normal aur acceptable** hain

Step 4 — Build and Verify (Claude Code Khud Karta Hai)

Tests Likhta Hai → Server Start Karta Hai → Real Tool Calls Karta Hai → Server Band Karta Hai → Results Report

MCP Inspector: Ek visual tool hai jo tools ko agent integration se *pehle* akele test karne deta hai — bahut useful for debugging!

> *"Composing clear tool descriptions is the actual challenge — harder than writing server code itself."*
> *— Emma*

5

Connect and Test from WhatsApp

OpenClaw se Connect Karo aur Test Karo

Step 1: Server Start Karo

```
uv run python server.py
```

Expected output: `Uvicorn running on http://127.0.0.1:8000`

- Terminal **khula rehna chahiye** — band karo ge toh server band ho jaega
- **Do alag terminal windows** use karo — ek server ke liye, ek OpenClaw commands ke liye

Step 2: OpenClaw Se Connect Karo

```
openclaw mcp set tutorclaw
'{"url":"http://localhost:8000/mcp","transport":"streamable-http"}' openclaw
gateway restart
```

Chapter 56 wala hi pattern hai — sirf URL ab locally built server ki taraf point kar raha hai.

Step 3: Registration Check Karo

```
openclaw mcp list openclaw mcp show tutorclaw
```

⚠️ **Warning:** URL mein ek bhi character galat ho toh — `/mcp` missing, wrong port — **silent failure** hogi, koi error message nahi aayega!

Step 4: WhatsApp Se Test Karo

Yeh message bhejo:

```
Register a new learner named James Chen
```

Agent ka expected behavior:

- Message ko Lesson 4 ki tool description se **match kare**

- Learner name ke saath **tool call kare**
- Learner ID ke saath **welcome message return kare**

Expected response: "*Welcome, James Chen! Your learner ID is 5262ec54-8d3d-49fe-8ccd-5ede5b9d173f.*"

Step 5: Tool Execution Verify Karo

- Dashboard mein conversation log dhundho
- Response ke paas tool badge dikhni chahiye: `register_learner`

Sirf text response kaafi nahi kyun? Agent plausible-sounding answer generate kar sakta hai bina actual tool call kiye — **tool badge hi prove karti hai** ke pura chain kaam kiya!

Complete Chain (Verification)

WhatsApp Message → Agent Tool Select → MCP Server Execute → Gateway → WhatsApp Response

★ Chapter 57 Ka Khulaasa

Summary of Key Learnings

Jo James (Aur Hum) Ne Seekha:

- Agent ke paas **knowledge** hoti hai — magar **ability** nahi, bina tools ke
- **Consumer vs Producer** — ab hum producer ban gaye hain
- **mcp-builder skill** — Claude Code ko domain expert banata hai
- **CLAUDE.md** — project rules aur workflow define karta hai
- **Streamable HTTP stateless** — best transport for network deployment
- **Describe-Steer-Verify cycle** — tools banane ka structured tarika
- **Tool description** — sabse zaroori element — agent isi se decide karta hai
- **Tool badge** — proof ke tool actually execute hua

> *"James ka reaction: Warehouse connections mein weeks lagte the aur consultants chahiye hote the — yahan itna jaldi kaam ho gaya!"*
> *"Emma: Pattern same rehta hai chahe tool type kuch bhi ho."*

❓ MCQs — 15 Questions

1 James ke agent ki kya main limitation thi jab student ne progress check karne ko kaha?

- **A** Agent ke paas internet access nahi tha
- **B** Agent training data se jaanta tha, magar live systems se interact nahi kar sakta tha ✓ Correct
- **C** Agent bahut slow tha
- **D** Agent sirf English mein kaam karta tha

2 MCP mein "Consumer" aur "Producer" ka kya farq hai?

- **A** Consumer code likhta hai, Producer sirf configure karta hai
- **B** Consumer existing server use karta hai, Producer apna naya server banata hai ✓ Correct
- **C** Consumer free hota hai, Producer paid hota hai
- **D** Consumer Python use karta hai, Producer TypeScript

3 mcp-builder skill install karne ka sahi command kaunsa hai?

A npm install mcp-builder

B pip install mcp-builder

C npx skills add https://github.com/anthropics/skills --skill mcp-builder ✓ Correct

D openclaw skill add mcp-builder

4 CLAUDE.md mein "spec-first development" rule kyun add kiya gaya?

A Code faster likhne ke liye

B Python use karne ke liye

C Pehle requirements agree karne ke liye — taake code likhne se pehle sab clear ho aur rework na ho ✓ Correct

D Server automatically restart karne ke liye

5 Teen MCP transport options mein se local-only operation ke liye kaunsa best hai?

A Streamable HTTP

B SSE (Server-Sent Events)

C stdio ✓ Correct

D WebSocket

6 James ne apne TutorClaw MCP server ke liye kaunsa transport choose kiya aur kyun?

A stdio — kyunke yeh sabse fast hai

B SSE — kyunke yeh events ke liye best hai

C Streamable HTTP stateless — network-accessible hai aur session management ki zaroorat nahi ✓ Correct

D Stateful HTTP — kyunke sessions yaad rakhna zaroori tha

7 Stateful transport ka kya nuksaan hai jo Emma ne apni story mein share kiya?

A Bahut expensive hota hai

B Sirf Python mein kaam karta hai

C Session cleanup, memory leaks aur mid-request expirations — raat 2 baje debugging karni padi ✓ Correct

D WhatsApp se connect nahi hota

8 Tool specification mein sabse zaroori element kaunsa hai aur kyun?

A Tool ka naam (lowercase)

B Tool ki description — agent isi se decide karta hai ke kab yeh tool call karna hai ✓ Correct

C Output format

D Input parameters ki quantity

9 Describe-Steer-Verify cycle mein "Steer" karne ka kya matlab hai?

A Server restart karna

B Claude Code ko band karna

C Spec ko modify karna — description sharpen karna, optional parameters add karna, output adjust karna ✓ Correct

D Test cases likhna

10 TutorClaw server start karne ke baad OpenClaw se connect karne ke liye pehla command kya hai?

A openclaw mcp connect tutorclaw

B openclaw mcp set tutorclaw '{"url":"http://localhost:8000/mcp","transport":"streamable-http"}' ✓ Correct

C openclaw server start tutorclaw

D uv run python server.py --connect

11 WhatsApp par tool execution verify karne ke liye sirf text response kaafi kyun nahi hai?

A Text response hamesha galat hota hai

B Agent plausible-sounding answer generate kar sakta hai bina actual tool call kiye — tool badge hi prove karti hai ✓ Correct

C WhatsApp text responses show nahi karta

D Dashboard mein text responses nahi dikhte

12 MCP Inspector ka kya kaam hai?

A Network security check karna

B Python code debug karna

C Tools ko agent integration se pehle akele visually test karne dena ✓ Correct

D WhatsApp messages inspect karna

13 openclaw mcp list ke baad agar tutorclaw nahi dikhta toh possible reason kya hai?

A Server bahut fast chal raha hai

B URL mein single character error ho sakti hai — jaise /mcp missing ya wrong port — jo silent failure deta hai ✓ Correct

C Python version galat hai

D Gateway properly installed nahi tha

14 mcp-builder skill ke SKILL.md file mein kya hota hai?

A Sirf project ka naam

B User ki profile information

C MCP protocol knowledge, SDK patterns, best practices aur testing approaches ✓ Correct

D Database connection strings

15 Training data aur tool mein fundamental farq kya hai jaise Emma ne explain kiya?

A Training data free hai, tools paid hote hain

B Training data theoretical knowledge deta hai (kya hota hai), tool actual ability deta hai (abhi live system par kya hai) ✓ Correct

C Training data zyada accurate hoti hai tools se

D Tools sirf Python mein kaam karte hain, training data kisi bhi language mein

---

Chapter 57: Extend Your Claw with MCP · Part 5: Building OpenClaw Apps
Agent Factory Curriculum · Notes & MCQs in Roman Urdu